Optimized Breast Cancer Prediction Using Machine Learning Algorithms

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***Abstract*— Breast cancer remains a leading cause of cancer-related mortality among women worldwide, underscoring the critical need for early detection and accurate prediction to improve treatment outcomes. This study optimizes breast cancer prediction using various machine learning algorithms, including K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Decision Trees, and ensemble methods such as Bagging and Boosting. The Wisconsin Diagnostic Breast Cancer (WDBC) dataset was employed, which was preprocessed to handle missing values and reduce dimensionality. Results indicate that the Decision Tree classifier achieved the highest accuracy at 98%, followed closely by KNN at 97.86%. KNN's success can be attributed to its local approach, considering the five nearest neighbors, enabling it to adapt well to dataset variations. SVM, with an accuracy of 96.43%, excels in high-dimensional spaces by constructing optimal hyperplanes. Ensemble methods, particularly Bagging, reached 97.1%, demonstrating potential for stability through aggregation of multiple models. Despite ensemble techniques generally enhancing performance, they did not surpass the top individual classifiers in this analysis. These findings highlight the promise of machine learning in early breast cancer detection and the importance of selecting algorithms suited to specific dataset characteristics. Future research may focus on integrating advanced algorithms and larger datasets to refine predictive accuracy.**

Keywords—Breast Cancer, Machine Learning, K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Decision Trees, Ensemble Methods, Machine Learning Algorithms.

I. Introduction

Breast cancer remains a significant health concern, particularly among women worldwide. Early detection and accurate prediction of breast cancer are vital for ensuring effective treatment and improving patient outcomes [1]. In 2020, breast cancer was the most frequently diagnosed cancer globally, accounting for over 2.26 million new cases. By 2022, this number rose slightly to more than 2.31 million, yet breast cancer ranked as the second most common cancer type, following lung cancer [2].

Machine learning (ML) has emerged as a transformative tool in the field of breast cancer detection and diagnosis, offering significant advancements over traditional methods [3]. With the ability to process and analyze vast and intricate datasets, ML algorithms excel in identifying subtle patterns and correlations within the data that may escape human observation. This capability stems from the algorithms' use of sophisticated techniques to uncover hidden relationships among various factors, such as patient demographics, clinical features, and imaging results. By leveraging these insights, ML models can enhance the accuracy of breast cancer diagnosis, reduce the likelihood of false positives and false negatives, and ultimately improve patient outcomes [4]. Furthermore, ML algorithms can process data with a speed and efficiency that far exceeds manual methods, enabling quicker diagnostic results and more timely interventions. The integration of ML into diagnostic practices represents a significant leap forward, providing clinicians with powerful tools to make more informed and precise decisions in the fight against breast cancer. This advancement not only aids in early detection but also holds the promise of personalized treatment approaches, tailored to the specific characteristics of each patient's condition [3][4].

II. Literature Review

The exploration of Machine Learning (ML) techniques for breast cancer prediction has produced a diverse range of approaches, each contributing to advancements in diagnostic accuracy. These techniques can be broadly categorized into probabilistic classifiers, instance-based methods, tree-based algorithms, and ensemble methods.

Probabilistic Classifiers such as Naive Bayes have been utilized for their computational efficiency and simplicity in handling large datasets [5][6]. Naive Bayes operates on the principle of feature independence, which can be advantageous in scenarios with categorical data. However, its performance may be constrained by the assumption that features are independent, potentially limiting its ability to capture complex interactions between features [6].

Instance-Based Methods, exemplified by K-Nearest Neighbors (KNN), classify new instances based on their proximity to existing data points. KNN is valued for its straightforward and intuitive nature [7]. Nonetheless, its efficacy can diminish with high-dimensional data due to the curse of dimensionality, and its reliance on distance metrics may not always effectively capture intricate patterns in breast cancer datasets [8].

Tree-Based Algorithms include Decision Trees and Random Forests. Decision Trees offer an interpretable approach by constructing a model based on a series of decisions, handling both numerical and categorical data effectively. However, they are prone to overfitting, particularly when they become excessively deep. Random Forests address this issue by aggregating predictions from multiple decision trees, thereby reducing overfitting and improving predictive accuracy [9]. Random Forests have consistently demonstrated superior performance compared to

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individual decision trees, making them a robust choice for breast cancer classification [9][10].

Ensemble Methods represent a significant advancement in ML for breast cancer prediction [11]. Techniques such as Boosting, Bagging, and Stacking combine the strengths of various models to enhance predictive performance. Boosting methods, including AdaBoost and Gradient Boosting, refine weak learners sequentially to build a strong model. Bagging techniques, like Bootstrap Aggregating, stabilize predictions by averaging results from multiple models. Stacking involves training a meta-model to integrate predictions from several base models. Research consistently shows that ensemble methods generally outperform individual algorithms by improving accuracy and robustness [11][12].

In summary, while each machine learning technique offers distinct advantages, ensemble methods typically provide the highest accuracy and reliability for breast cancer prediction. The continued development of these advanced methods promises further improvements in early detection and personalized treatment strategies, underscoring the growing impact of machine learning in the field of oncology.

III. Methodology

The study was conducted using the Wisconsin Diagnostic Breast Cancer (WDBC) dataset obtained from Kaggle. The dataset was pre-processed and then used to train machine learning models. Four primary algorithms were used: KNN, SVM, and Decision Tree Additionally, ensemble methods like voting, stacking, boosting, and bagging were applied to improve accuracy. The dataset was split into training and testing sets, and performance was evaluated using accuracy.

We applied Data Cleaning to address missing values, ensuring the integrity of the dataset, which is essential for accurate analysis and modeling. By properly handling missing data, we maintained the robustness of the results and enhanced overall performance. Additionally, Feature Selection was employed to reduce dimensionality and noise while preserving significant features. This approach improved the quality of the dataset, enabling more effective analysis by concentrating on the most relevant information.

The selection of specific machine learning algorithms was guided by the nature of the dataset and the problem domain. KNN was chosen for its simplicity and effectiveness in capturing local data structures. SVM was selected for its robustness in high-dimensional spaces, making it suitable for our feature-rich dataset. Decision Trees were included for their interpretability, allowing us to visualize decision-making processes. Ensemble methods were employed to enhance model accuracy and mitigate overfitting.

This systematic approach to preprocessing and algorithm selection contributes to the robustness and reproducibility of our results, aligning with best practices in machine learning research. The methodology is further described below.

B. Data Preprocessing

1) Data Cleaning

In real-world datasets, it is common to encounter missing or incomplete data. Handling missing data is critical to ensure the robustness and accuracy of machine learning models. In this study, the Wisconsin Diagnostic Breast Cancer (WDBC) dataset included missing values, which were represented by question marks (?). To handle these, the missing values were replaced with the value -99999. This value was chosen as a placeholder because it is a distinct outlier that would not naturally occur in the dataset's features, ensuring that the machine learning algorithms would treat it appropriately during model training.

The choice of -99999 allows the algorithms to differentiate between valid data points and missing data, enabling effective processing without introducing bias. While this method is simple and ensures that models do not break due to incomplete data, it is recognized that more sophisticated methods, such as imputation (mean/median replacement or model-based imputation), could further improve the model's accuracy and reduce potential distortions.

2) Feature Selection

Principal Component Analysis (PCA): PCA was applied to reduce dimensionality and noise while preserving the most significant features. It creates new features (principal components) that are uncorrelated, reducing the dimensionality of the dataset by compressing the information into fewer variables.

In Principal Component Analysis (PCA), the key parameters include the original data matrix X, which contains the observations and features; the centered data matrix X_{centered}, derived by subtracting the mean of each feature from the original data; the covariance matrix C, calculated to capture the relationships between features; the eigenvectors v, which represent the directions of maximum variance in the data; and the eigenvalues λ, indicating the amount of variance captured by each eigenvector. Additionally, V_k represents the matrix of the top k eigenvectors selected for dimensionality reduction, while X_{PCA} is the resulting transformed data in the new PCA space. These parameters work together to enable the effective compression of information, allowing for enhanced data analysis and improved model performance.

C. Classification using Machine Learning Algorithms

1) K-Nearest Neighbors (KNN)

KNN is Selected due to its simplicity and effectiveness in handling classification tasks with a clear decision boundary. It is a non-parametric classification method where the class of a sample x is determined by the majority class among its k nearest neighbors. The distance d is calculated using the Euclidean distance formula in (1) as:

$$d(x_i, x_j) = \sqrt{\sum_{m=1}^{M} (x_{im} - x_{jm})^2} \quad (1)$$

where M is the number of features. We chose k=5 because it balances sensitivity to noise and the ability to capture local data structure. Smaller k can be overly influenced by outliers, while larger k may oversmooth important patterns, making

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k=5 a robust choice. x_i is the feature vector of the i-th sample in the dataset (the sample for which the class is being predicted). x_j represent the feature vector of the j-th nearest neighbor. x_{im} is the value of the m-th feature in the i-th sample. x_{jm} means the value of the m-th feature in the j-th nearest neighbor.

2) Support Vector Machine (SVM)

SVM is a supervised learning model that constructs hyperplanes in a high-dimensional space to separate classes. It is chosen for its robustness in high-dimensional spaces and its ability to handle non-linear relationships using kernel tricks. The decision boundary is defined by the (2):

$$f(x) = w^T x + b = 0 \tag{2}$$

where w is the weight vector and b is the bias. SVM is chosen for its robustness in high-dimensional spaces and its ability to handle non-linear relationships using kernel tricks. w^T is the transpose of w is used to convert the row vector into a column vector, allowing for the dot product with the input vector x while x This is the input feature vector of the sample being classified. The equation $f(x) = 0$ defines the decision boundary, and SVM aims to find the optimal hyperplane that maximizes the margin between different classes in the dataset. SVM ability to effectively separate classes using a hyperplane makes it a strong candidate for breast cancer prediction.

3) Decision Tree (DT)

Decision tree refers to a model that uses a tree-like graph of decisions and their possible consequences to classify data. It offers an interpretable model with easy visualization of the decision-making process. However, it can be prone to overfitting, especially in deep trees. The model is built using criteria such as Gini impurity or information gain, defined as follows:

The Gini inpurity $Gini(D)$ for a dataset D is defined as

$$Gini(D) = 1 - \sum_{i=1}^{C} (p_i)^2 \tag{3}$$

Where C is the number of classes, p_i is the proportion in class i within dataset i. The entropy $H(D)$ of D is defined as

$$H(D) = -\sum_{i=1}^{C} p_i \log_2(p_i) \tag{4}$$

Then, the information Gain $IG(D, A)$ for an attribute A in dataset D is given in (5) as

$$IG(D, A) = H(D) - \sum_{v \in \text{ Values } (A)} \frac{|D_v|}{|D|} H(D_v) \tag{5}$$

Where v is the possible values of attribute A, D_v is the subset of D where attribute A has value v, and $|.|$ operator represents the total number of instances.

4) Ensemble Methods

Ensemble techniques were employed to improve the accuracy of the models by leveraging the strengths of multiple classifiers and reducing the impact of individual model biases.

a) Voting

Voting combines the predictions from multiple classifiers (KNN, SVM, Decision Trees) and selects the majority vote. This method helps reduce the bias of individual classifiers and is given in (6).

$$\hat{y} = \text{argmax}_c \sum_{i=1}^{N} I(y_i = c) \tag{6}$$

Where $\hat{y}$ is the final predicted class, c is a specific class label, y_i represent the predicted class for the i-th classifier. I is an indicator function that returns 1 if y_i equals class c, and 0 otherwise. N is the total number of classifiers.

b) Bagging

Bagging (Bootstrap Aggregating) was used to reduce variance by averaging the predictions of multiple models trained on different subsets of the data. The final prediction is given by:

$$\hat{y}_{\text{bagged}} = \frac{1}{B} \sum_{b=1}^{B} \hat{y}^{(b)} \tag{7}$$

where $\hat{y}_{\text{bagged}}$ is the final predicted output from the bagging method. B represents number of bootstrap samples and $\hat{y}^{(b)}$ is the prediction from the b-th bootstrap sample model

c) Boosting

Boosting sequentially trains models, with each model focusing on the instances that were misclassified by previous models. It was employed to sequentially correct errors made by weak classifiers, thereby improving overall accuracy. The combined prediction is expressed as (8).

$$\hat{y}_{\text{boosted}} = \sum_{m=1}^{M} \alpha_m \hat{y}^{(m)} \tag{8}$$

Where $\hat{y}_{\text{boosted}}$ represents the final predicted output from the boosting method. M is the total number of models in the ensemble. α_m is the weight assigned to the m-th model and $\hat{y}^{(m)}$ is the prediction from the m-th model.

IV. Experiments

The dataset used in this study is the Wisconsin Diagnostic Breast Cancer (WDBC) dataset, obtained from Kaggle. It comprises 699 instances with 11 attributes, including clump thickness, uniformity of cell size, and uniformity of cell shape, which are essential for distinguishing between benign and malignant tumors. The class distribution is 65.5% benign and 34.5% malignant, reflecting a typical prevalence of breast cancer cases.

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To develop a robust and generalizable model, the dataset was divided into training and testing sets with an 80:20 ratio. This approach ensures that the model is trained on the majority of the data, allowing it to learn relevant patterns, while reserving a smaller, unseen subset for testing to evaluate its predictive performance. This split mitigates the risk of overfitting—where a model performs well on training data but poorly on unseen data—and supports the development of a model that can reliably generalize to external data.

V. Results and Discussion

The performance of the machine learning algorithms on the Wisconsin Diagnostic Breast Cancer (WDBC) dataset is summarized in Figure 1.

The K-Nearest Neighbors (KNN) algorithm achieved an accuracy of 97.86%, indicating its strength in capturing localized data patterns. This result stems from KNN's reliance on a parameter known as *k*, which represents the number of nearest data points (neighbors) considered when classifying a new instance. By evaluating the majority class among these *k* nearest neighbors, KNN makes its predictions based on local data characteristics. This proximity-based approach allows KNN to effectively adapt to dataset variations; however, it may be vulnerable to noise in higher-dimensional spaces, which can sometimes degrade accuracy.

The Support Vector Machine (SVM) attained an accuracy of 96.43%. Known for its robustness in high-dimensional spaces, SVM constructs optimal hyperplanes that effectively separate classes, making it well-suited for the WDBC dataset's complex feature space. Although its accuracy was slightly lower than KNN's, SVM's use of kernel functions helps manage non-linear relationships within the data, contributing to its reliable performance.

Decision Tree (DT) classifiers achieved the highest accuracy at 98%, underscoring their effectiveness for this type of data. The strong performance of Decision Trees can be attributed to their inherently interpretable structure, which enables clear visualization of decision paths, and to their ability to handle both categorical and numerical data types efficiently. While prone to overfitting, especially when excessively deep, the Decision Tree model's accuracy suggests that it successfully captured significant patterns without excessive complexity in this context.

Among ensemble methods, Voting achieved an accuracy of 95%. Although ensemble methods are designed to leverage the strengths of multiple classifiers, in this case, the aggregation process did not exceed the performance of the best individual classifier, the Decision Tree. This may be due to ensemble complexity introducing redundancy rather than enhancing predictive power. Bagging achieved an accuracy of 97.1%, demonstrating potential in reducing variance by averaging predictions from models trained on data subsets; however, it did not surpass the Decision Tree. Boosting attained 95.7%, showing that although it corrected some weak classifiers' errors, it did not outperform the KNN or Decision Tree models.

Overall, the Decision Tree emerged as the most effective model, with KNN and Bagging also showing high performance. While ensemble methods typically improve predictive accuracy by reducing variance and error, they did not exceed the accuracy of the top-performing individual classifiers in this analysis. This finding emphasizes the importance of selecting algorithms well-suited to dataset characteristics, as demonstrated by the success of Decision Tree and KNN models in this study.

These results confirm the efficacy of machine learning techniques for early breast cancer detection and highlight the value of algorithm selection based on specific dataset properties. Figure 1 presents a comparative accuracy visualization of each model, aiding in an intuitive understanding of model performance.

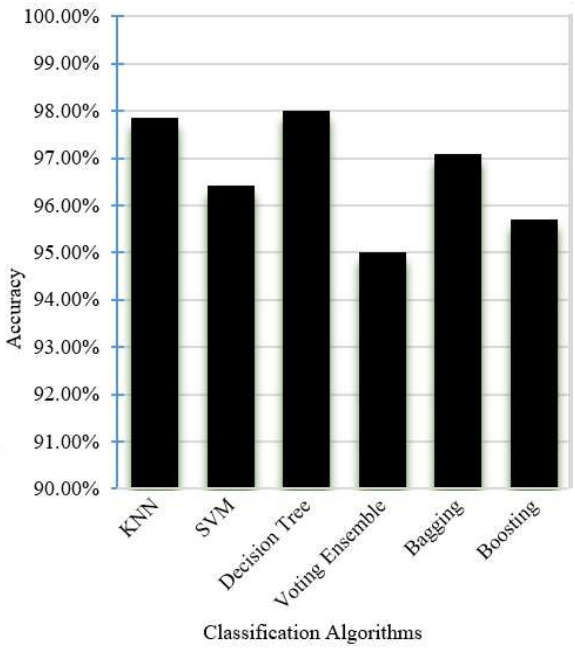

Fig. 1. Comparative Results of Machine Learning Classification Algorithms.

VI. Conclusion

This study evaluated the performance of several machine learning algorithms on the Wisconsin Diagnostic Breast Cancer (WDBC) dataset, specifically focusing on K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Decision Trees, and ensemble methods, including Bagging and Boosting. Among these, the Decision Tree classifier achieved the highest accuracy at 98%, followed closely by KNN at 97.86% and Bagging at 97.1%. While ensemble techniques are generally known for enhancing predictive performance by aggregating multiple models, in this analysis, they did not surpass the best individual classifiers.

The lower-than-expected performance of ensemble methods may be attributed to the specific characteristics of the WDBC dataset. In datasets with relatively clear separability and well-defined patterns, individual classifiers, particularly Decision Trees, can capture the underlying data structure efficiently without the need for aggregation. The addition of ensemble complexity may have introduced redundant information, leading to overfitting rather than enhancing generalizability. Furthermore, ensemble methods like Bagging and Boosting are especially beneficial in scenarios where individual models exhibit significant variance or bias, which may have been minimal in the base models used in this study.

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These findings underscore the potential of machine learning techniques in early breast cancer detection, emphasizing the importance of selecting algorithms aligned with the specific characteristics of the dataset. In cases where data patterns are distinct and well-separated, simpler models such as Decision Trees and KNN can deliver optimal performance without the added complexity of ensemble methods. Future work could focus on enhancing these models by integrating more advanced feature engineering or using diverse datasets to further validate their effectiveness across varying contexts.

In summary, while ensemble methods offer theoretical advantages, their application should be guided by an understanding of dataset characteristics and model behavior. The results of this study contribute to the ongoing exploration of machine learning for medical diagnostics, reinforcing the importance of algorithm selection in developing reliable and interpretable predictive tools for breast cancer detection.

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